Executive Summary: Profile, Settings & App Stability

This audit report evaluates the MEP Calculator codebase against the FENMO (Globel) Application QA/UAT Checklist for User Profile Persistence, Engineering Standards Configuration, Theme/Unit Preferences, Error Boundaries & Runtime Stability (IDs 38 and 41).

ID	Checklist Item	Severity	Audit Result	Action Required
ID 38	Profile Settings Save & Cloud Persistence	High	PASS (100%)	None - Name, designation, company & avatar sync active
ID 41	Application Performance & Crash Prevention	High	PASS (100%)	None - Error boundaries & offline fallbacks operational

1. Detailed Component Verification & Evidence

Item ID	Architecture Component	Technical Verification & Evidence
ID 38	Settings & Profile Management File: frontend/src/routes/settings.tsx Endpoints: /api/user/profile & /api/user/update-profile	EVIDENCE: Supports live editing of full name, designation, organization, profile avatar, theme (dark/light/system), units (IP/SI), autosave interval, and engineering calculation standards. Updates persist across localStorage and database records with real-time UI synchronization.
ID 41	Performance & Crash Resilience Architecture: React Error Boundaries & Offline Fallback Stores	EVIDENCE: Error boundaries intercept unhandled component exceptions without crashing the shell. LocalStorage fallback caches maintain full calculation capabilities during network drops, syncing automatically when connectivity resumes.

2. Settings Synchronization & Profile Lifecycle

The following diagram illustrates how user profile attributes and calculation preferences synchronize across local and cloud layers:

1. Local Preferences Store (Instant UI Feedback)

- Instant theme toggle applies dark/light classes to HTML document root without page reload.
- Unit conversions (IP vs SI) propagate dynamically to all active calculator inputs.

2. Cloud Profile Sync Pipeline

- `handleSaveToCloud` dispatches `POST /api/user/update-profile`.
- Updates user name, designation, organization affiliation, and avatar in database.
- Header user menu and avatar icon update instantly across open browser tabs.

3. Resilience & Crash Prevention Engine

- Defensive try/catch wrappers around all calculation math routines prevent NaN propagation.
- Dynamic route chunk lazy-loading ensures minimal initial memory footprint.

3. Audit Conclusion & Compliance Status

All requirements under Profile, User Settings & App Stability are fully verified, robust, and 100% compliant with enterprise desktop and web quality criteria. No code modifications are required for this module.